

Customer Lifetime Value (LTV) Prediction Model

1. Introduction

Customer Lifetime Value (LTV) is a strategic metric used to estimate the total revenue a customer is expected to generate throughout their relationship with a business. Accurate LTV estimation enables organizations to optimize marketing spend, improve customer retention strategies, and enhance long-term profitability. In data-driven digital ecosystems such as digital wallet platforms and e-commerce systems, predictive analytics plays a critical role in identifying high-value customers early.

2. Abstract

This project develops a machine learning-based regression model to predict Customer Lifetime Value using historical transaction data. Transaction-level data was aggregated at the customer level and transformed into behavioral features including Recency, Frequency, and Average Order Value (AOV). An XGBoost regression algorithm was trained to model non-linear relationships between customer behavior and revenue generation. Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 score. The predicted LTV values were further used to segment customers into value-based categories to support targeted marketing and strategic decision-making.

3. Tools Used

- Python – Core programming environment for data processing and modeling.
- Pandas & NumPy – Data manipulation and numerical computations.
- Scikit-learn – Data splitting and evaluation metrics.
- XGBoost – Gradient boosting regression model for LTV prediction.
- Matplotlib – Feature importance visualization.
- Google Sheets – Dashboard creation and business reporting.

4. Steps Involved in Building the Project

Step 1: Data Preprocessing – The dataset was cleaned by removing duplicates and handling missing values. Transaction dates were converted to datetime format for time-based aggregation.

Step 2: Feature Engineering – RFM-based features were generated: Recency (days since last purchase), Frequency (number of transactions), Total Spend, and Average Order Value (AOV).

Step 3: Model Training – The dataset was split into training and testing sets. An XGBoost regression model was trained to predict customer-level revenue using engineered features.

Step 4: Model Evaluation – Performance was assessed using MAE, RMSE, and R^2 metrics to ensure prediction accuracy and robustness.

Step 5: Customer Segmentation – Predicted LTV values were categorized into Low, Medium, and High-value segments using percentile thresholds.

Step 6: Dashboard Integration – Final outputs were exported to Google Sheets for KPI creation, pivot analysis, and visualization.

5. Conclusion

The project demonstrates the practical application of machine learning in customer analytics. By leveraging behavioral features and gradient boosting regression, the model provides accurate and scalable LTV predictions. The integration of predictive outputs with customer segmentation enables data-driven marketing, optimized budget allocation, and improved customer retention strategies. This approach supports strategic planning by transforming raw transactional data into actionable business intelligence.